

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application Based Question

Course Code:	CSA5110	Course Title:	Cryptography and Network Security
CO Assessed:	CO1-Imitation (BL3)	Assessment:	Assessment Tool 1– Application Based Question
Weightage:	40% of CIA	Total Marks:	25
CO1 Statement:	<i>Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)</i>		
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Section / Batch:		Date:	16-07-2026

Code of Conduct

I T.Sanjeeva Reddy - 192372304 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- **This assessment contains 5 questions.**
- **Each question carries 10 marks. Total: 50 Marks.**
- **Ensure clarity, structure, and logical flow in answer**
- **Duration: 50 Minutes.**

Question s	Understandi ng of Concepts(25)	Experimenta l design(30)	Use of Tools(20)	Analysi s of Result (15)	Documentatio n (10)	Total Mark s
Overall Total (100)						

Question 1: Security Attacks

Incident 1: Fraudulent Emails Asking for Login Details

Attack Name: Phishing Attack

Attack Category: Active Attack

Description

Phishing is a social engineering technique where attackers impersonate trusted organizations by sending fake emails or creating fraudulent websites to trick users into revealing sensitive information such as usernames, passwords, or banking credentials.

Impact on the CIA Triad

Confidentiality

Confidential customer information, including login credentials, may be exposed to attackers.

Integrity

Once unauthorized access is obtained, attackers can alter account details or perform unauthorized financial transactions.

Availability

Legitimate users may lose access to their accounts if passwords are changed or accounts are compromised.

Preventive Measures

Enable Multi-Factor Authentication (MFA).

Deploy advanced email filtering solutions.

Conduct regular cybersecurity awareness programs for employees and customers.

Use spam detection systems.

Implement secure email gateways to identify malicious messages.

Incident 2: Unauthorized Transactions Using Stolen Session Tokens

Attack Name: Session Hijacking

Attack Category: Active Attack

Description

Session hijacking occurs when an attacker captures a valid session token and uses it to impersonate an authenticated user, allowing unauthorized access to banking services.

Impact on the CIA Triad

Confidentiality

Sensitive session information and user data may be exposed.

Integrity

Attackers can perform unauthorized fund transfers or modify banking records.

Availability

Genuine users may be logged out or temporarily prevented from accessing their accounts.

Preventive Measures

Enforce HTTPS with TLS encryption.

Use Secure and HttpOnly cookie attributes.

Configure automatic session expiration after inactivity.

Generate new session tokens after successful login.

Use VPNs for secure communication over untrusted networks.

Incident 3: Excessive Traffic Targeting the Banking Server

Attack Name: Distributed Denial-of-Service (DDoS) Attack

Attack Category: Active Attack

Description

In a DDoS attack, multiple compromised devices simultaneously send a large volume of requests to a server, making banking services unavailable to legitimate users.

Impact on the CIA Triad

Confidentiality

There is generally no direct impact on confidential information.

Integrity

The stored banking data remains unchanged.

Availability

Banking services become slow or completely inaccessible due to server overload.

Preventive Measures

Configure network firewalls.

Implement Intrusion Prevention Systems (IPS).

Use load balancing to distribute traffic.

Subscribe to DDoS mitigation services.

Apply traffic rate limiting to block abnormal requests.

Incident 4: Employees Using Weak Passwords

Attack Name: Password Attack (Brute Force/Dictionary Attack)

Attack Category: Active Attack

Description

Attackers use automated software to repeatedly guess passwords until the correct one is found. Weak or commonly used passwords significantly increase the risk of unauthorized access.

Impact on the CIA Triad

Confidentiality

Confidential employee information and organizational data may be exposed.

Integrity

Unauthorized users can modify or delete sensitive information.

Availability

Employee accounts may become locked or misused, interrupting normal operations.

Preventive Measures

Enforce strong password creation policies.

Store passwords securely using hashing algorithms such as bcrypt or Argon2.

Enable Multi-Factor Authentication (MFA).

Encourage the use of password managers.

Configure account lockout mechanisms after repeated failed login attempts.

Question 2: Security Services and Security Mechanisms

Requirement 1: Only Authorized Doctors Should Access Patient Records

Security Services

Authentication

Access Control

Recommended Security Mechanisms

Multi-Factor Authentication (MFA)

Role-Based Access Control (RBAC)

Biometric Authentication

Justification

Authentication confirms the identity of healthcare professionals before granting access. RBAC ensures that only authorized doctors can view or update patient records according to their assigned roles, while biometric verification provides an additional layer of security.

Requirement 2: Patient Records Must Not Be Altered Without Permission

Security Service

Integrity

Recommended Security Mechanisms

SHA-256 Hashing

Digital Signatures

Justification

SHA-256 generates a unique hash value for every record, making unauthorized modifications easy to detect. Digital signatures verify both the authenticity of the sender and the integrity of the medical data.

Requirement 3: Every Medical Transaction Should Be Traceable

Security Service

Non-Repudiation

Recommended Security Mechanisms

Digital Signatures

Audit Logs

Justification

Digital signatures ensure that healthcare professionals cannot deny performing authorized actions. Audit logs maintain a complete record of system activities, allowing every transaction to be tracked for accountability and compliance.

Requirement 4: Patient Data Shared Between Hospitals Must Remain Confidential

Security Service

Confidentiality

Recommended Security Mechanisms

AES-256 Encryption

TLS/SSL Protocol

Virtual Private Network (VPN)

Justification

Strong encryption protects patient information while it is transmitted across networks. TLS/SSL secures communication channels, and VPNs create encrypted connections that prevent unauthorized interception of sensitive medical data.

Requirement 5: The EHR System Should Continue Operating During Cyberattacks

Security Service

Availability

Recommended Security Mechanisms

Firewall

Intrusion Detection System (IDS)

Intrusion Prevention System (IPS)

Backup Servers

DDoS Protection

Justification

These mechanisms work together to defend the healthcare infrastructure from cyber threats, detect and block malicious activities, and maintain uninterrupted access to critical medical services even during attacks.

Question 3: Caesar Cipher

Given Ciphertext

YMNX NX F XJHWJY RJXXFLJ

Step 1: Identify the Encryption Key

The ciphertext was decrypted using the brute-force method by testing all possible shift values. The correct key was found to be a shift of 5.

Encryption Key = 5

Step 2: Decryption Process

Applying a backward shift of 5 letters to each character:

$Y \rightarrow T$

$M \rightarrow H$

$N \rightarrow I$

$X \rightarrow S$

Continuing the same process for the remaining characters gives the final plaintext:

Plaintext: "THIS IS A SECRET MESSAGE"

Step 3: Why the Caesar Cipher Is Not Secure

Although the Caesar Cipher is simple to understand, it provides very little security because:

It has only 25 possible shift keys, making it easy to crack.

Attackers can quickly recover the plaintext using brute-force attacks.

It is vulnerable to frequency analysis, where letter occurrence patterns reveal the original message.

Since each letter is always replaced by the same shifted letter, it is a monoalphabetic substitution cipher, making it predictable.

It does not provide adequate protection for modern digital communication.

Step 4: Recommended Encryption Technique

Advanced Encryption Standard (AES-256)

AES-256 is a modern symmetric encryption algorithm that offers significantly stronger security than the Caesar Cipher.

Advantages of AES-256

Uses a 256-bit encryption key, making it highly resistant to attacks.

Computationally impractical to break using brute-force methods.

Recognized as an international encryption standard.

Widely used to secure sensitive information in banking, government, military, healthcare, and cloud computing applications.

Provides strong confidentiality and is suitable for protecting modern digital data.

Question 4: Playfair Cipher

Keyword

DEFENCE

Step 1: Constructing the Playfair Matrix

Using the keyword DEFENCE (removing duplicate letters and combining I and J), the 5×5 matrix is formed as follows:

D	E	F	N	S
A	B	C	G	H
I	K	L	M	O
P	Q	R	T	U
V	W	X	Y	Z

Step 2: Ciphertext

BMODZBXDNABEKUDMUIXMMOUIVIF

Step 3: Decrypted Plaintext

After applying the Playfair decryption rules to each pair of letters, the original message is obtained as:

"HIDE THE GOLD IN THE TREE STUMP"

Advantages of the Playfair Cipher Compared to the Caesar Cipher

The Playfair Cipher offers better security than the Caesar Cipher for several reasons:

- It encrypts pairs of letters (digraphs) instead of individual characters.
- This approach reduces the effectiveness of simple frequency analysis.
- The use of a keyword-based matrix creates a much larger key space.
- It provides stronger protection than a basic monoalphabetic substitution cipher.

Limitations of the Playfair Cipher

Despite being more secure than the Caesar Cipher, the Playfair Cipher still has several weaknesses:

- It is susceptible to digraph frequency analysis, which can reveal patterns in encrypted text.
- Modern cryptanalysis techniques can break the cipher with sufficient ciphertext.
- It does not provide authentication to verify the sender's identity.
- It cannot ensure data integrity or detect message tampering.

- **Due to these limitations, it is not suitable for securing modern military, financial, or online communications.**